

UPMANYU JHA

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WORK EXPERIENCE

Quantiphi Analytics Solutions Private Limited (Google Cloud Partner)

Mumbai, IN

Machine Learning Engineer (LLMs, NLP, Computer Vision, Agents & RAG)

Jul 2023 - Present

Scale & Architecture: Refactored document processing into scalable microservices handling 80K+ documents quarterly, boosting classification accuracy by 19.5% and reducing manual review time by 80% through targeted model retrofitting and pipeline optimization

- Engineered a financial anomaly detection model for transactions integrity using statistical profiling, increasing recall by 35% and cutting manual review queues by 50% while maintaining high precision
- Slashed model hallucinations by 85% via domain-specific fine-tuning and reduced GCP costs by 15% through strategic resource optimization
- Achieved 92% F1-score in structured data extraction using few-shot prompting, reducing manual processing time by over 40%

Leadership: Led a cross-functional squad of 6+ engineers, standardized production-grade AI model deployment across 5+ business units with reusable custom artifacts; cut ramp-up time by 20% and improved team delivery scores by 15%

- Deployed 15+ generative modeling pipelines on Google Cloud to automate processing for 1-2K daily documents, enabling scalable ingestion of unstructured data

Delivery: Architected 3-month technical roadmaps for 10+ projects, resolving scalability and reusability bottlenecks and accelerating AI/ML lifecycle delivery by 2-3 weeks

- Designed a Retrieval-Augmented Generation (RAG) pipeline with fine-tuned LLMs and GCP ETL services, elevating financial document extraction F1-score from 0.55 to 0.90 (+63% improvement)
- Engineered scalable MLOps and CI/CD pipelines using pytest to automate code reviews, increasing unit test coverage to 95% while cutting production bugs by 80% and deployment failures by 50%

Bugcrowd | Hackerone

Mumbai, IN

Security Researcher | Bug Bounty Hunter (Web Application, API)

Jun 2021 - Jul 2023

- Conducted Root Cause Analysis (RCA) on 30+ architectural vulnerabilities, scripting automated compliance checks to prevent recurrence
- Developed Python/Docker automation tools to identify security gaps, boosting manual testing efficiency by 60%

TECHNICAL SKILLS

- Programming Languages:** Python, Go, SQL (PostgreSQL), GraphQL, C++, C, Git, Bash, **Tools:** Linux, Github, Flask, Django, Postman, Jira, Rally
- Machine Learning:** PyTorch, TensorFlow, Keras, TensorRT, XGBoost, OpenCV, Scikit-Learn, **Hugging Face** (Transformers: BERT, LSTM)
- Visualization:** Matplotlib, Seaborn, Plotly, **Data & ML Infrastructure:** Pandas, NumPy, MLflow, Airflow, DVC, **VectorDBs** (Pinecone, FAISS)
- Generative AI & LLMs:** OpenAI API, Gemini, Anthropic (Claude Opus), **Fine-tuning** (LoRA), **Agents** (n8n, LangChain, crewAI)
- Cloud & MLOps:** GCP (Doc AI, ADK, Vertex AI, GKE, Storage, BigQuery), **Amazon Web Services** (SageMaker, S3, EC2, RDS), Docker, Kubernetes

PROJECTS AND RESEARCH

Smart Traffic Management System (Research Prototype) | GANs, Reinforcement Learning (RL) and Computer Vision (CV)

Dec 2025 - Present

- Developing an AI-agent traffic control prototype integrating Object Detection (YOLO) and multi-agent Reinforcement Learning (RL) in SUMO simulation to target a 10% reduction in commute time

AstraShield - Disaster Relief Verification System | Python, FastAPI, Gemini Vision AI, React, GCP

Oct 2025 - Nov 2025

- Engineered Forensic Agentic AI achieving a 90% detection rate via Human-in-the-Loop workflow across 100+ real-world samples, utilizing Gemini Vision Pro and Tesseract OCR for multimodal analysis
- Architected real-time pipeline integrating Google Custom Search and WHOIS, delivering background checks and risk scores in under 30 seconds
- Built scalable FastAPI backend implementing Model Context Protocol (MCP) for Agentic function calling with evidence logging; deployed with React frontend for safe donation verification

Alphafold - Drug Discovery (Healthcare/Pharma) | DeepMind AlphaFold

Jul 2023 - Jan 2024

- Won Think-Tank (H1-24) Award** for leveraging AlphaFold in a healthcare POC, advancing protein and peptide structure prediction accuracy for a pharmaceutical client's Phase-1 trial

LEADERSHIP AND OPEN SOURCE CONTRIBUTION

Teaching Assistant | Mentor - HAWK-i Labs

Aug 2021 - Present

- Co-led Python / C++ / NLP workshops for 1000+ students, significantly improving preparedness for 12th Board and University exams

Project - iSee: Drafted Lightweight, privacy-first Apple utility shared on Swift subreddit; achieved 17.4% engagement among 46k+ viewers

Blogging: Authored technical articles on Medium (hackergod00001), gaining 1,000+ followers and establishing authority with 20,000+ audience

EDUCATION

University Of Mumbai

Mumbai, IN

Bachelor of Engineering in Computer Engineering, **GPA: 8.92/10** | **Relevant Coursework:** Deep Learning, Computer Vision, Cloud Computing (AWS)

CERTIFICATIONS AND TRAINING

Google Generative AI Leader (**GCP Gen AI**) | Google Cloud

Dec 2025

Google Cloud Professional Cloud Architect (**GCP PCA**) | Google Cloud

Jun 2025

Google Cloud Professional Machine Learning Engineer (**GCP PMLE**) | Google Cloud

May 2024